

Artefaux v2 — Discard / Extreme Stress Pack: Explainer

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2026-07-16

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This is the plain-language companion to the v2 waveform review pack ([reports/artefaux_v2_ecg_review.pdf](#), rendered with every lead as a full 10-second strip). v2 is deliberately **weighted toward clinically-unusable recordings**: **50 of 85** records are expected to be discarded. It exists to stress a signal-quality gate where it matters — near and beyond the point where an ECG stops being interpretable.

Scope. A *stress-test set*, not a *clinically representative cohort*. It over-samples failure on purpose; do not read prevalence or deployment accuracy from it.

1. The three groups at a glance

The corpus is 85 stress records in three groups, each with its own colour throughout this document and the figures.

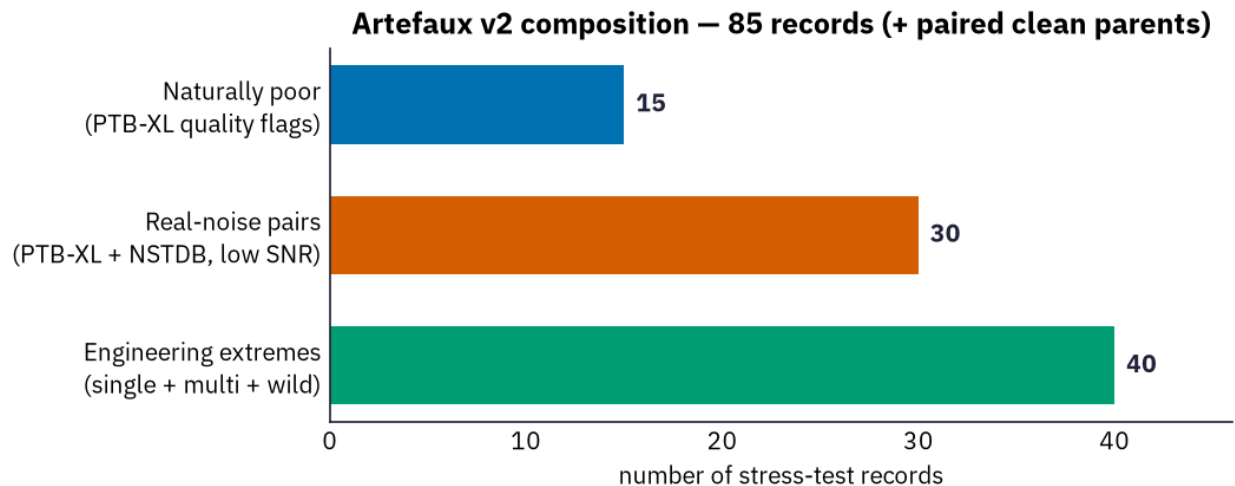


Figure 1: Composition of the v2 corpus by group.

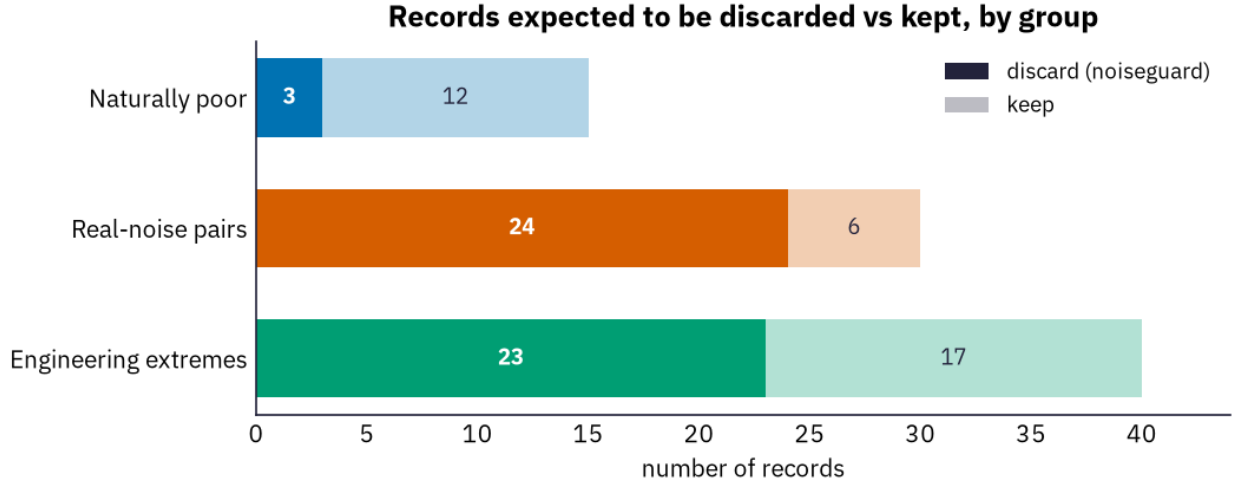


Figure 2: How many records each group expects a gate to discard versus keep.

Group	n	Discard	Reject	Integrity failures	Source	Corruption
Naturally poor	15	3	3	0	PTB-XL (quality flags)	none (inherently noisy)
Real-noise (low-SNR)	30	24	2	0	clean PTB-XL + NSTDB	em/ma/bw @ {-6, -4, -2, +0, +2} dB
Engineering extremes	40	23	5	22	clean PTB-XL (+ MACECGDB)	deterministic acquisition failures
Total	85	50	10	22		

2. What each group is for

Naturally poor (15) — real-world noisy reference. Genuine PTB-XL 12-lead ECGs flagged by PTB-XL’s own technical-validation quality columns (`static_noise`, `burst_noise`, `baseline_drift`, `electrodes_problems`). No synthetic corruption and no clean parent — this is what real acquisition noise looks like, and it anchors the engineered cases against reality. The severe electrode-problem records are labelled reject/discard; the rest are **limited** but genuinely noisy.

Real-noise (30) — a controlled low-SNR dose-response. Clean PTB-XL parents mixed with real MIT-BIH NSTDB noise (`em` electrode motion, `ma` muscle artefact, `bw` baseline wander) at a fixed, *aggressive-low* SNR ladder {-6, -4, -2, +0, +2} dB. Unlike v1’s clean-to-noisy ladder, v2 drops the easy end and concentrates on the accept/reject decision region, so a detector’s behaviour can be traced as a function of severity rather than measured at a single point. Each ships with its untouched clean parent (`*_clean`) for paired comparison.

Engineering extremes (40) — deterministic acquisition failures. Recipe-built breakage rather than added noise: rail-clipping saturation, flat / stuck-constant / disconnected / digital-missing (NaN) channels, huge electrode-motion swings, step-and-recovery baseline shifts, polarity reversal, intermittent lead-off, and compound “wild” multi-mode records. Single-lead, multi-lead, and whole-precordium variants exercise both per-lead and record-level gate logic.

3. How labels are assigned

Every record carries three label layers: the **clinical parent** (authored rhythm class + PTB-XL quality flags), the **corruption truth** (exact leads/electrodes, artefact, requested + measured SNR, seed, amplitude bookkeeping), and the **expected behaviour** (what a correctly-tuned gate should do). Labels are authored from the recipe — never by running a detector.

Expected behaviour is written in **two deliberately different vocabularies**: **signalguard** (a graded record verdict — **diagnostic** / **limited** / **rhythm_only** / **reject**, plus a per-lead good/borderline/bad) and **noiseguard** (a binary record-level *discard*). A record can be **limited** for one and **discard=false** for the other; the label captures both.

Real-noise: SNR → label. For whole-record contamination the bands are (generated from the definition, so they match the corpus exactly):

SNR (dB)	Record quality	Lead quality	Discard
-6	reject	bad	yes
-4	limited	bad	yes
-2	limited	bad	yes
+0	limited	borderline	yes
+2	limited	borderline	—

Subset (limb- or chest-only) contamination degrades only the touched half, so the record stays **limited** and discards only at or below 0 dB.

Naturally poor. Severe electrode-problem records are **reject** / **discard**; the remaining quality-flagged records are **limited** and kept.

Engineering. Labels are authored per failure mode from what the recipe actually does to each lead. Crucially, **NaN / flat / stuck-constant / rail-saturated cases are labelled data_integrity_failure with a specific integrity_failure_type — not as noise** — because they are acquisition/software faults a gate must catch by a different route than a low SNR.

4. The waveform pack

The companion `reports/artefaux_v2_ecg_review.pdf` renders each record with **every independent lead (I, II, V1–V6) as its own full 10-second strip** (25 mm/s, 10 mm/mV, diagnostic filter), so a reviewer sees each lead’s complete recording at once and rail-clipping shows honestly as a **clipped** marker. Regenerate the whole set with:

```
make regenerate                                     # build out/artefaux-v2 from local
python scripts/reports/render_ecg_review.py --layout strip8 # -> reports/artefaux_v2_ecg_review
python scripts/reports/generate_v2_explainer.py --pdf      # this document
```

*Generated by Artefaux v2.0.0 (<https://github.com/depolarised/artefaux>) via `scripts/reports/generate_v2_expl`
every count and the SNR→label table are computed from the committed corpus definition.*